

Functional prototype Sprint plan for Biodex

Sprint duration: 2 weeks – week 7-9

Sprint goal: to create a functional, high-quality prototype through multiple iterations

Sprint start: 7/09/2026

Sprint end: 20/09/2026

Sprint schedule:

Time	Main Activity	Team Activity
Day: 2–3	Core development	Iterate over medium fidelity design and discuss areas of improvement.
Days 4–5	Brainstorming / starting OOP design	Ensure the app follows and satisfies the features as defined in user stories and start OOP design.
Days 6–8	Refine frontend and develop backend.	Develop frontend while improving the medium fidelity feedback. Develop flow critical components that must be finished for the rest of the group to proceed.
Day 9	Main copy of project done	Aim to finish project development
Day 10-14	Refinement	Feedback and refine project, address any errors.

1. Project & Application Foundation

Responsible: Team effort

The first checkpoint is to build the foundational piece for everyone to work on, such as the SQL structure, JFX project structure. The team needs to apply the theme agreement on titles, naming convention, dimensions, navigation bar, animations, UI components from the medium fidelity checkpoint.

Should have completed from previous checkpoint:

- Set up the JavaFX project.
- Configure SQLite/JDBC.
- Establish Git/repository structure.
- Create shared navigation/scene routing.
- Establish basic application styling.

- Create the initial database structure.
- Agree on shared components such as buttons, forms, cards and error messages.

Sprint output: To iterate on the baseline and develop rich features.

2. Login & Signup

Responsible: Ali

Ali will develop the Login/Signup specification, outline specification requirements, and decide which specs to include. Medium fidelity had Login and Signup screens, form validation. Successful authentication is important because other sections such as settings and upload rely on knowing which user is currently signed in.

Tasks:

- Build a Login screen.
- Build Signup screen.
- Add username/email and password fields.
- Connect authentication to the database.
- Create the user session.
- Add navigation to Forgot Password and 2FA.
- Implement logout.
- Ample feedback to users.

Dependency: Important because other sections are reliant of this section.

Sprint output: A user can sign up, log in and reach the main application.

3. Forgot Password & 2FA

Responsible: Forgot Password: Vinny; 2FA: Bernard

Support: Ali for Login integration

Aim to develop by day 8. These features can be developed along with the login/sign-up because they are all similar flows. The team members should communicate so the flow is consistent across the three screens. 2FA implementation was deemed difficult so a default value of “111111” was added as a place hold. The plans to have a six-digit email code, expiry handling, resend and a back-up button in case authentication fail was scrapped.

Tasks:

- Create Forgot Password screens.
- Separate group meeting.
- Create verification-code interface.
- Create a reset-password interface.
- Add navigation between the three stages.
- Create a 2FA six-digit verification screen.
- Add expiry/error states.
- Add resend functionality.
- Create the flow so the three screens are connected.

Dependency: Ali needs to provide the Login navigation points, while Vinny + Bernard need to maintain the shared password/verification rules.

Sprint output: A user can access password recovery or 2FA from the Login flow and return to the application.

4. Species Search & Species Details

Responsible: Vinny

Covers the main search functionality that allows the user to find invasive species, while species Details displays information about the selected species and its local sightings. This page should be adjusted according to the technical constraints of the project. The main flow should be focused instead of making sure every part of the project is functional.

Tasks:

- Build Species Search screen.
- Add search/filter interface, display species results, create Species Details screen.
- Display species image and information.
- Add a threat profile.
- Display local sightings.
- Add "Log Sighting".
- Add "Add Photo".
- Add "Open Full Map".

Sprint output: A user can search for a species, open its details and move into reporting/map functionality.

5. Upload / Report Sighting

Responsible: Dong Hyeon

The Upload section is where users can upload individual species and update sightings, severity or provide feedback/fix mistakes. The section should have a coherent upload + submission page with ample user feedback. It should have checks to ensure there is only a maximum of five files uploaded and a 10mb limit. The specification requires users to upload an image and provide information such as location, species, date, notes and severity. It will all be stored locally, with no database connection.

Tasks:

- Create an upload/report screen.
- Add file picker.
- Add image preview.
- Validate required information.
- Add submission confirmation.
- Store the basic sighting information.
- Robust feedback to user
- Robust backend and injection protection
- Seamless integration and flow to severity heat map and profile

Dependency: The upload page is dependant on the authentication sections and the homepage.

Sprint output: A robust backend development and user upload feature.

6. Heat Map

Responsible: Joshua

The heatmap section adds visual feedback of various sightings across Brisbane. It should be Brisbane centred, with various filters for time and species and a button for the report sighting section that will take users to the upload page. The scope might exceed the scope for this checkpoint, so the team should get a basic map working and focus on good design.

Tasks:

- get the map designed, api linked
- Display sample/initial sighting locations with emphasis on simplicity.
- Add species filter, date filter, map legend, sighting interaction.
- Connect selected sightings to Species Details.
- Add Report Sighting button.

Dependency: To be developed simultaneously with the upload page. Therefore, the first sprint can use sample data while the database integration is being completed.

Sprint output: A working map interface showing invasive-species sightings and basic filtering.

7. User profile

Responsible: Yash

The settings page would have limited functionality but should have a coherent design and have dark mode / light mode working. All settings are user-scoped, persisted in a relational database, and loaded once per session into memory for fast UI binding. The team should develop the page so that its functional.

Tasks:

- Add buttons for settings such as user features, design etc.
- Add profile.
- Add customisation (light/dark mode).
- Add my uploads check.
- View and edit account details.

Dependency: Require the login to be completed beforehand, will sit alongside user upload, view threats. Will not be accessible for non-logged-in users.

Sprint output: Light/dark mode does not have to be implemented immediately, nor does the display of the user profile. Main focus should be on the flow and UI, compactness.

Sprint completion:

The sprint plan can be concluded when the team has produced a high-quality medium-fidelity prototype of Biodex with the features outlined below.

Satisfactory:

Working clickable navigation between all major screens

Search/filter interaction

Species cards/results
Detailed species information
Threat-level indicators
Local sightings
Map with realistic sample sightings
Upload screen with image preview
Report form with validation states
Capable flow for login, forgot and reset password
Basic profile/settings screens
Confirmation after submission
2FA screen as a prototype interaction

Minimum requirements:

AREA	MINIMUM REQUIREMENT
NAVIGATION	Good design with transition between pages
LOGIN	Login screen exists with text input, but no verification or hashing implemented
HOME/SEARCH	Good user design where user can search for species.
REPORT SIGHTING	Users can upload and report sightings
SPECIES DETAILS	Users can view information about species
MAP	Users can view heatmaps
CONSISTENT UI	Cohherent design philosophy throughout
MAIN INTERACTION	Buttons, forms, areas working and properly implemented
ERROR STATES	Basic levels of user feedback for forbidden attempts / graceful failing